SMART CENTER PIVOT IRRIGATION & MONITORING SYSTEM

Alec Anderson, Jack Pipkin, and Jackson Flynn

OBJECTIVE & GOALS

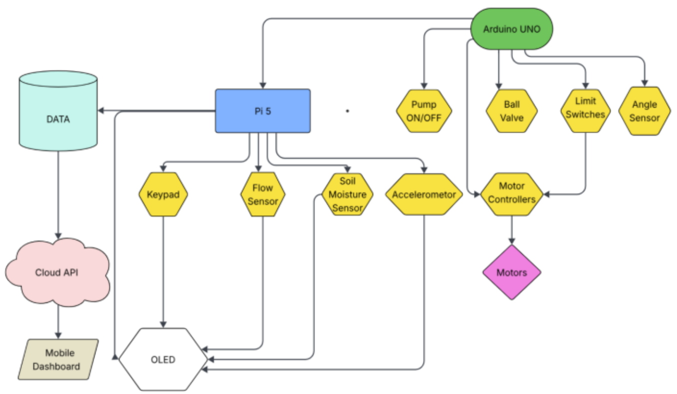
- Design a working irrigation demonstration platform
- Integrate water control and motion control into one system
- Use sensor inputs to support automated operation
- Provide local operator control through a keypad and OLED interface
- Create a platform that can support data logging, cloud upload, and a cellphone dashboard

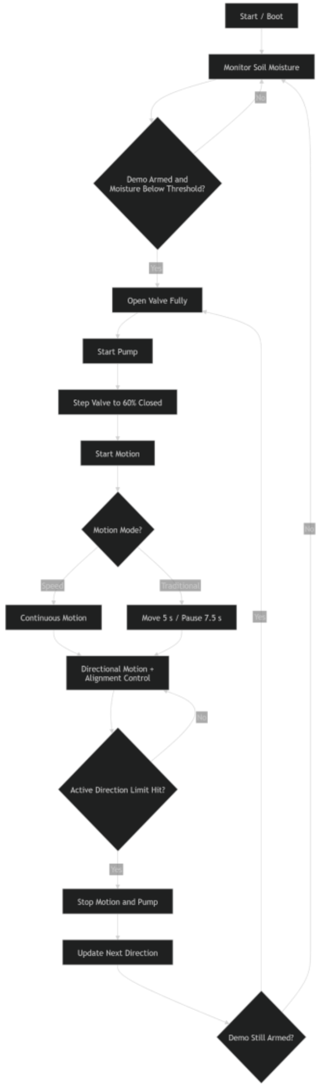
SYSTEM OVERVIEW

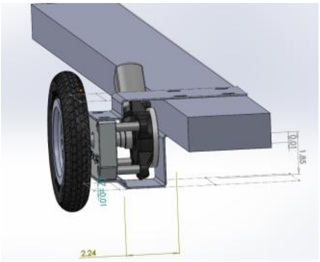
The system is built around a **Raspberry Pi** and an **Arduino Nano**.

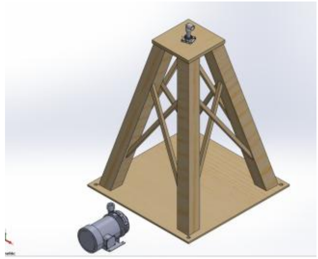
- The **Raspberry Pi** handles the user interface, local data handling, keypad input, OLED display, and higher-level supervision.
- The **Arduino Nano** handles real-time local control for motion and water operation.
- Sensors provide information on soil moisture, flow, vibration, alignment, and travel limits.
- Pump relays and ball valve relays control water delivery.
- Motor control logic allows the system to move in either direction while maintaining alignment.

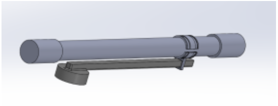
SYSTEM DESIGN AND LOGIC FLOW



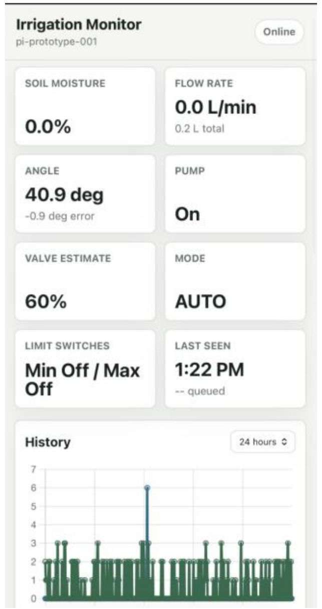








MOBILE DASHBOARD & OLED



OLED / Keypad Interface

- Live display of soil moisture, flow, vibration, and controller status
- Manual pump and ball valve control
- Drive control with speed, direction, and motion mode selection
- Demo mode arming, threshold adjustment, and status display

CURRENT CAPABILITIES

The current prototype successfully demonstrates:

- local control of pump, ball valve, and motion
- sensor-based irrigation sequence triggering
- directional limit switch behavior
- automatic demo rearming
- selectable motion modes
- local user interface through keypad and OLED
- cloud logging and cellphone dashboard